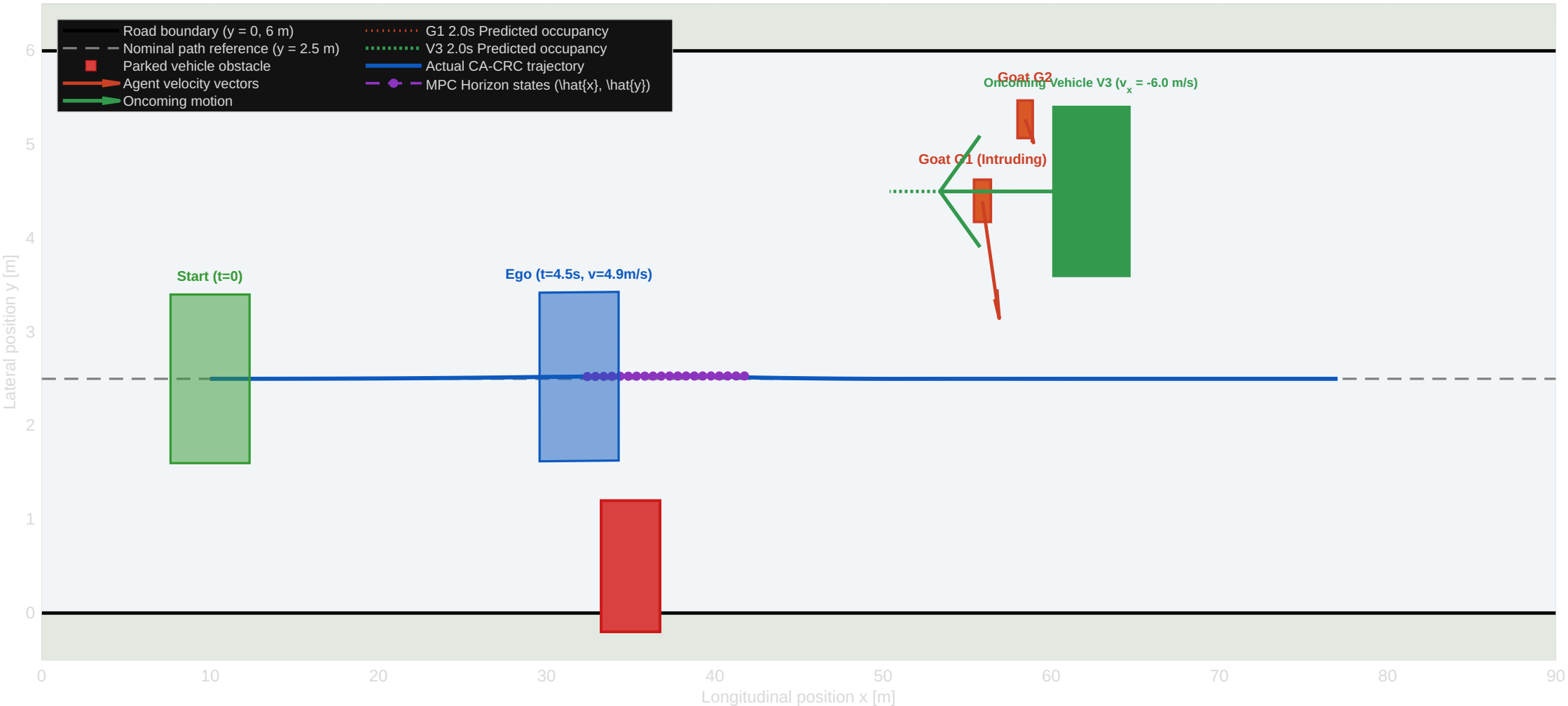


MOMENT OF DECISION (t = 4.5s): Temporal Multi-Agent Bottleneck Conflict & Safety-Filtered Yielding



(a) Perception & Agent Tracking State

Simulation Time: t = 4.5 s (Step 46)
Ego State: x=31.9m, y=2.52m, v=4.87m/s
Tracked Entities: 3 Dynamic + 1 Static

Agent ID	Type	x [m]	y [m]	vx [m/s]	vy [m/s]
G1	Livestock	55.9	4.39	0.20	-0.25
G2	Livestock	58.5	5.27	0.10	-0.05
V3	Vehicle	62.4	4.50	-6.00	0.00

(b) Free-Space & Physical Clearance Audit

Physical Corridor Parameters:
Road Boundary Limits: $y \in [0.0\text{m}, 6.0\text{m}]$
Minimum Predicted Clearance: 2.07 m
Available Corridor Width: 6.00 m
Relative Closing Speed: 10.9 m/s

Temporal Feasibility Check:
Left Corridor ($y > 4.0\text{m}$): BLOCKED by Oncoming V3
Right Corridor ($y < 2.0\text{m}$): BLOCKED by Intruding G1

(c) CA-CRC Planner & Safety Filter Audit

MPC Optimization Output:
Status Code: 1 (FEASIBLE / HARD QP)
Unconstrained u_{MPC} : $[\delta = -0.001 \text{ rad}, a = 0.01 \text{ m/s}^2]$

Safety Filter Monitoring:
Safety Filter Status: PASS (Constraints satisfied)
Executed Control u_{safe} : $[\delta = -0.001 \text{ rad}, a = 0.01 \text{ m/s}^2]$

Decision Action:
TEMPORAL YIELD & DECELERATE ($v \rightarrow 2.0 \text{ m/s}$)
Wait for Oncoming V3 Clearance before Lateral Evasion